

# Incentivizing Retail Traders: Evidence from Daily High-Water Marks on a Social Trading Platform

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# First a little background

- Anthony and I have known each other for years
  - Visiting Illinois alum Filipe Correia at UGA
- This summer I joined this paper
  - Expanding dataset to tick data
  - Doing a bit more microstructure
- Still finalizing the data collection; additional results at the end

# Technology is Changing How Retail Investors Trade

- Fintech has changed how consumers interact with financial markets
  - Spending (D'Acunto et al. 2019b) and saving (Gargano and Rossi 2024)
  - Robo-advisors help investors diversify (D'Acunto et al. 2019a)
- Platform design shapes the decisions retail investors make
  - 'Gamified' interfaces alter investor attention (Barber et al. 2022; Tan 2021)
- *A growing and relatively unexplored innovation: social trading platforms*

# Social Trading: Retail Traders as Fund Managers

- Retail **traders** make their investment decisions public
  - Retail **investors** buy an exchange-traded product that tracks the trader's portfolio
  - Retail traders effectively become fund managers
- The platform pays traders a performance bonus
  - A fraction of the fund's return above its previous high close
  - Earned any day the portfolio closes above that high, and the threshold resets immediately
- *An institutional high water mark contract, but paid daily to retail traders* (Ma et al. 2019; Aragon and Nanda 2012)

# How Could Daily Incentives Affect Retail Traders?

- Alignment
  - Performance pay aligns trader and investor interests, potentially mitigating retail biases
    - (Jensen and Meckling 1976; Holmstrom 1979)
- Or amplification
  - Extreme frequency amplifies loss aversion, excessive trading, and short-term thinking
    - (Odean 1998, 1999; Dorn et al. 2015; Delfabbro et al. 2021)
- **This paper: How does a daily high water mark bonus change the behavior and performance of retail traders?**

# Data and Methodology

- Data
  - **Social trading platform:** every trade, daily fund prices, and bonus days from 2018 to 2022
  - **Sample:** 5,147 funds and 3.6 million fund-day observations
  - **CRSP and LSEG:** returns of the underlying assets
  - *Increasing our coverage with German exchange data*
- Methodology
  - Regression discontinuity around the bonus threshold
  - Differences-in-differences around each bonus (de Chaisemartin and D'Haultfoeuille 2026)
  - January 1st reset of the high water mark
  - Counterfactual static portfolio where traders do not trade at all

# Preview of the Main Results

- On days a bonus is achieved
  - Returns spike 1.6 percentage points above the previous day
    - Immediately fall back to pre-bonus levels
  - Odds of trading are 15% higher just above the bonus threshold
  - Traders sell recent winners to lock in the gain, but miss further run up
- Doing nothing would be better: 63% of funds would earn higher alpha and 57% of fund-years would earn more bonuses
- Consistent with daily incentives amplifying short-term focus rather than aligning interests

# Contribution

- Performance incentives are well studied for institutional managers (Brown et al. 1996; Massa and Patgiri 2009; Agarwal et al. 2009; Ray 2009); pay can align interests (Jensen and Meckling 1976; Holmstrom 1979)
  - Daily HWM amplifies short-term focus among retail traders
- Fintech platform design can help (D'Acunto et al. 2019a; Rossi and Utkus 2021; Gargano and Rossi 2024) or hurt users (Barber et al. 2022; Tan 2021)
  - Wall Street style incentives backfire at a daily frequency
- Retail and social trading (Barber and Odean 2000; Petajisto 2013; Anginer et al. 2024; Oehler et al. 2016; Roder and Walter 2019; Oehler and Schneider 2023)
  - Retail–professional gap persists even with performance pay

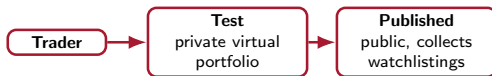
# Social Trading (Wikifolios)

- Individuals register and build a virtual portfolio using over ~600,000 financial products, including
  - Stocks
  - Exchange-traded products
  - Funds (real estate, bonds, equities, money market, or mixed funds)
  - Investment certificates (index certificates, bonus certificates, discount certificates, and reverse convertibles)
  - Leverage products
- Traders trade virtually, but prices reflect real-time bid/ask quotes
- An index is created to replicate the virtual portfolio's performance 1:1

# Starting a Wikifolio

- Any registered user can create a wikifolio: a virtual portfolio plus a written trading idea (at least 200 characters)
- The trader chooses the investment universe the wikifolio can trade
- **Test** status: private, only the trader can see it
- To publish: trader profile, title, idea, at least one trade, and verified email
- **Published** status: searchable, users can add it to their watchlist and follow every trade

# How a Wikifolio Works



## Becoming Investable

- At least 10 watchlistings and the wikifolio is at least 21 days old
- Legitimation: ID, address, and phone number (18 or older)
- Issuance call with wikifolio about the trading idea and strategy
- Lang & Schwarz issues an open-ended index certificate with its own ISIN that tracks the wikifolio 1:1
- Once issued, the investment universe is fixed

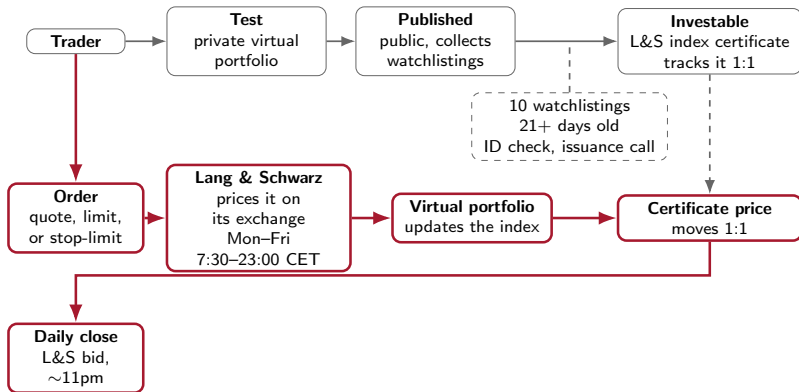
# How a Wikifolio Works



# Trading

- Trades are virtual, priced at real-time quotes for the underlying asset on the exchange run by Lang & Schwarz
  - Quote orders (price valid for 3 seconds), limit orders, and stop-limit orders
- Trading hours (CET): Mon–Fri 7:30am–11pm, Sat 10am–1pm, Sun 5–7pm
- No transaction costs for trades inside the wikifolio
- With invested capital, Lang & Schwarz hedges the positions
- Daily close = Lang & Schwarz bid price at about 11pm

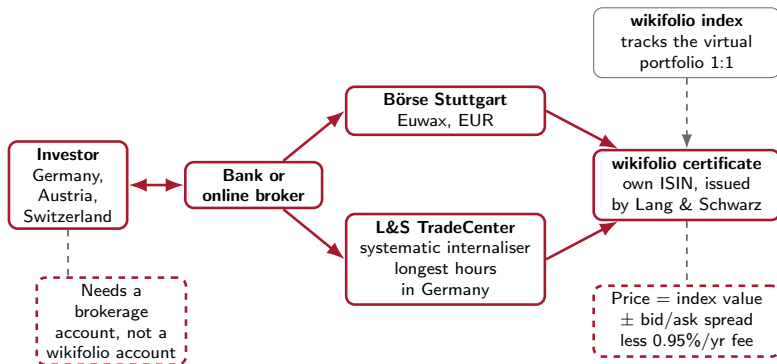
# How a Wikifolio Works



# Investing in Wikifolio Certificates

- Investors buy the certificate by its ISIN through their own bank or online broker, with no wikifolio account needed
- The certificate is not traded on the Lang & Schwarz exchange, but on
  - Börse Stuttgart (Euwx, in EUR), or
  - Directly with Lang & Schwarz TradeCenter: a systematic internaliser, Germany's off-exchange venue with the longest trading hours
- Who can buy
  - Germany: every certificate is publicly offered
  - Austria: certificates of Austrian traders or with high demand
  - Switzerland: only stock-only wikifolios (BX Swiss, in CHF)
- Price tracks the wikifolio 1:1, with a bid/ask spread from its holdings and a 0.95% annual certificate fee charged daily

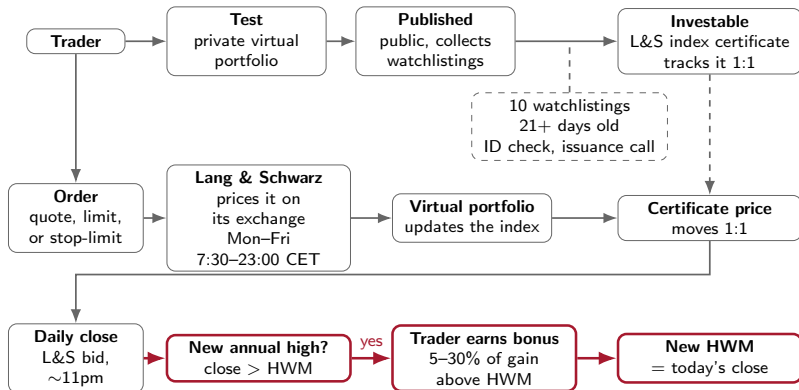
# How Investors Buy In



# Fees

- Traders earn no fee on assets under management, only performance bonuses
- Bonus rate of 5–30% of the gain above the high water mark (HWM), set by the trader at launch and fixed
- Earned only when the daily close sets a new annual high
- The HWM resets every January 1st

# How a Wikifolio Works



## Bonus Calculation

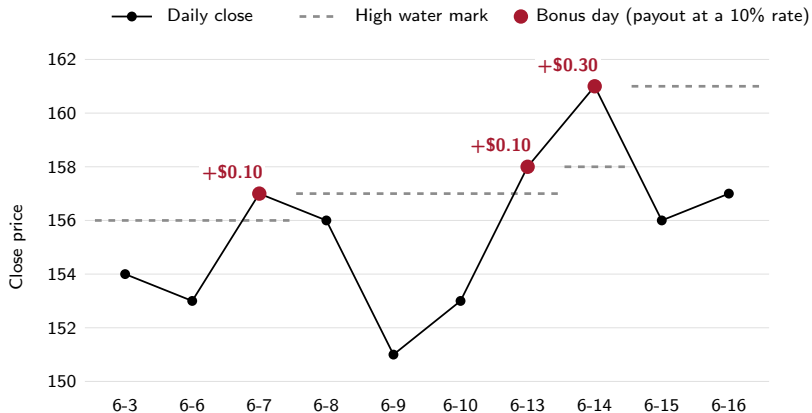
$$Bonus_{i,t} = \begin{cases} (Price_{i,t} - HWM_{i,t}) * Rate_i, & \text{if } Price_{i,t} > HWM_{i,t} \\ 0, & \text{otherwise} \end{cases}$$

## Sample High Water Mark

| Date    | Close Price | High Water Mark | Bonus Earned? |
|---------|-------------|-----------------|---------------|
| 6/3/22  | 154         | 156             | No            |
| 6/6/22  | 153         | 156             | No            |
| 6/7/22  | 157         | 156             | Yes           |
| 6/8/22  | 156         | 157             | No            |
| 6/9/22  | 151         | 157             | No            |
| 6/10/22 | 153         | 157             | No            |
| 6/13/22 | 158         | 157             | Yes           |
| 6/14/22 | 161         | 158             | Yes           |
| 6/15/22 | 156         | 161             | No            |
| 6/16/22 | 157         | 161             | No            |

Close Price = Lang & Schwarz bid at ~11pm CET

# Sample High Water Mark



$$\text{Bonus} = (\text{Close} - \text{HWM}) \times \text{Rate}$$

# Data

|                    | Current                          | Extension                                                                        |
|--------------------|----------------------------------|----------------------------------------------------------------------------------|
| Period             | 2018–2022                        | Through 2026                                                                     |
| Underlying returns | Daily returns from CRSP and LSEG | Tick-level exchange and OTC tape data, quotes and trades (finalizing collection) |

- **Filters:** drop closes below \$1, each fund's first month, January, and days over \$100 from a bonus; keep funds with 20+ trades a year and 6+ active months
- **Current sample:** 5,147 funds and 3.6 million fund-days (from 8,155 funds)

# Portfolio Characteristics

|                                     | Mean     | Median   | Standard Deviation |
|-------------------------------------|----------|----------|--------------------|
| Daily Trades                        | 1.17     | 0        | 4.94               |
| Daily Buys                          | 0.66     | 0        | 3.02               |
| Daily Sales                         | 0.51     | 0        | 2.49               |
| Daily Trades when Trades > 0        | 5.43     | 3        | 9.48               |
| Daily Quantity when Trades > 0      | 5,206    | 100      | 349,310            |
| Daily Dollar Amount when Trades > 0 | \$24,150 | \$2,086  | \$436,231          |
| Portfolio Life (Years)              | 2.65     | 2.37     | 1.49               |
| Daily Returns                       | 0.02%    | 0%       | 2.05%              |
| Daily Alpha                         | -0.0002% | -0.0001% | 0.37%              |
| Performance Fee                     | 10.78%   | 10%      | 6.29%              |
| Bonuses Earned Per Year             | 30.1     | 20       | 32.6               |

Fund size is set by the trader at creation

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Two snapshots of open wikifolios

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67% of funds exhibit a negative alpha

# Portfolio Characteristics

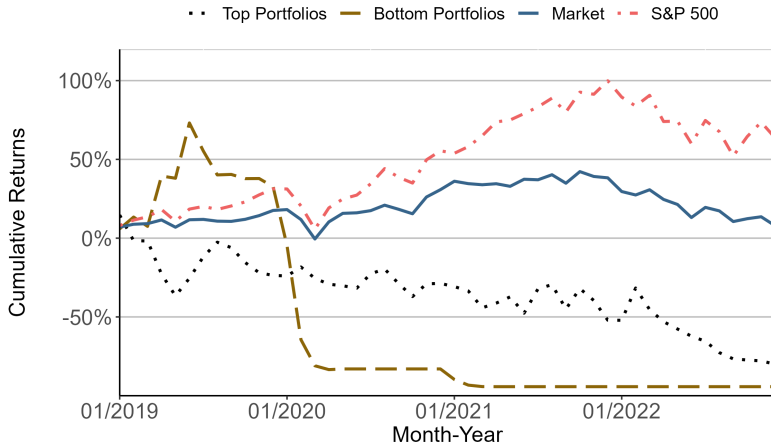
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Chosen by the trader, ranging from 5% to 30%

# Return Benchmarking

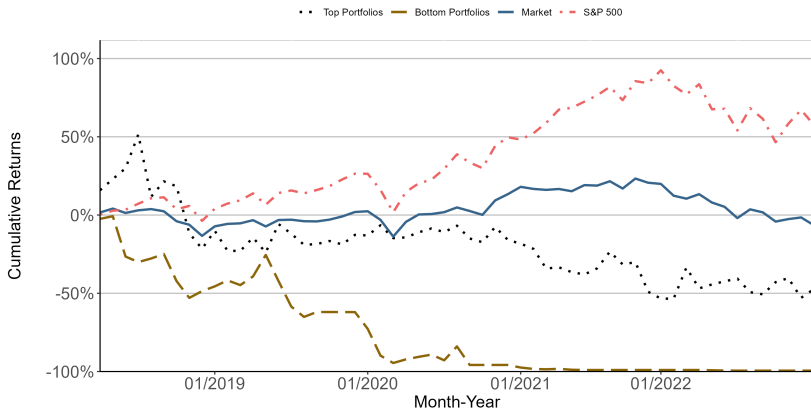
- Exercise
  - Rank funds by average return over the prior period
  - Invest in the ranked funds and hold for one period
  - Re-rank and rebalance either yearly, quarterly, or monthly
- Portfolios compared
  - **Top Portfolios:** 10 funds with the highest prior-period return
  - **Bottom Portfolios:** 10 funds with the lowest prior-period return
  - **Market:** equal-weighted, every fund active that period
  - **S&P 500:** benchmark, since 68% of funds trade US assets

# Yearly Rebalancing



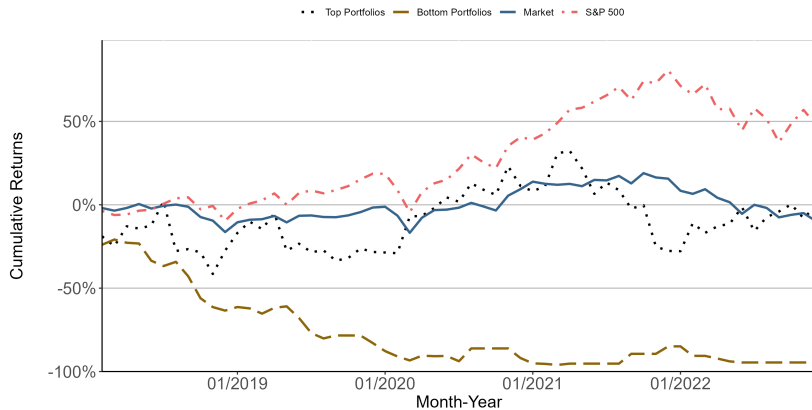
Both the top and bottom funds yield almost complete losses

# Quarterly Rebalancing



Slight improvement for the top funds, which only lose about 50%

# Monthly Rebalancing

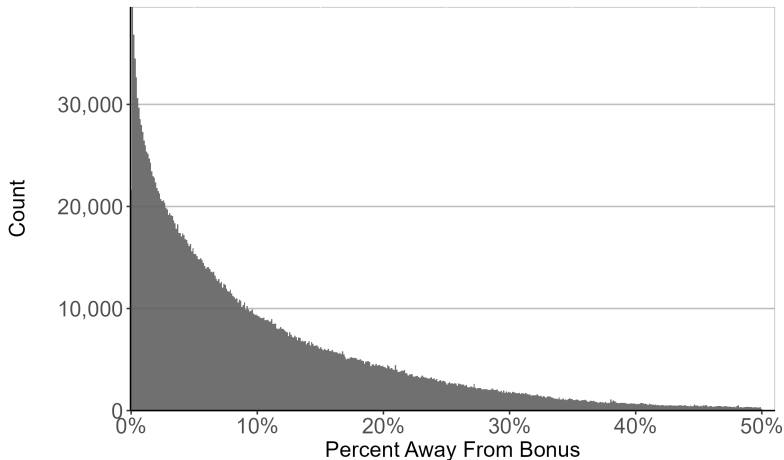


Top performers break even, but no consistent performance

# Why Doesn't Performance Persist?

- In line with traders favoring short-term strategies, rather than buy and hold
  - Most traded: leveraged ETFs on market indices, and stocks like Apple, Tesla, Amazon, and Nvidia
- Daily HWM could encourage this
- **How close are traders to a bonus on a typical day?**

## Distribution of Distance to Bonus

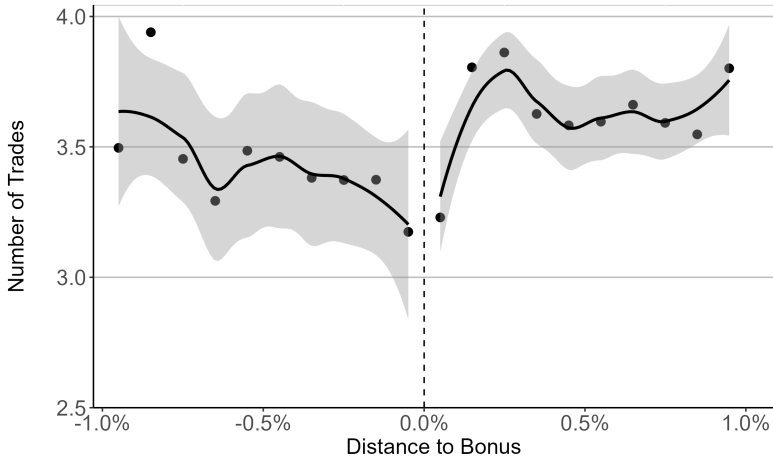


39% of all trader-days are within 5% of the bonus threshold

# Is the Bonus Salient?

- Regression discontinuity around the bonus threshold
- Compare trader-days just above vs. just below the threshold (within  $\pm 1\%$ )
- Outcomes over the prior 3 trading days (traders average about 1 trade a day):
  - Number of trades
  - Whether any trade was made

## RDD Around Bonus

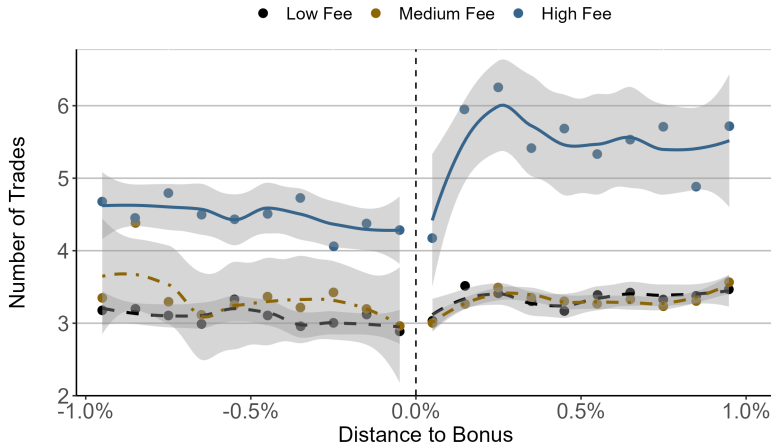


Intensive: 0.31 trades (9% of mean) | Extensive: odds increase 15%

# Bonus Fee Groups

- The trader picks a 5–30% bonus rate when creating the fund, and it cannot be changed
- Three groups:
  - Low:  $\leq 7.5\%$  (1,881 traders)
  - Medium: 7.5–15% (2,375 traders)
  - High:  $> 15\%$  (891 traders)

## RDD Around Bonus By Group



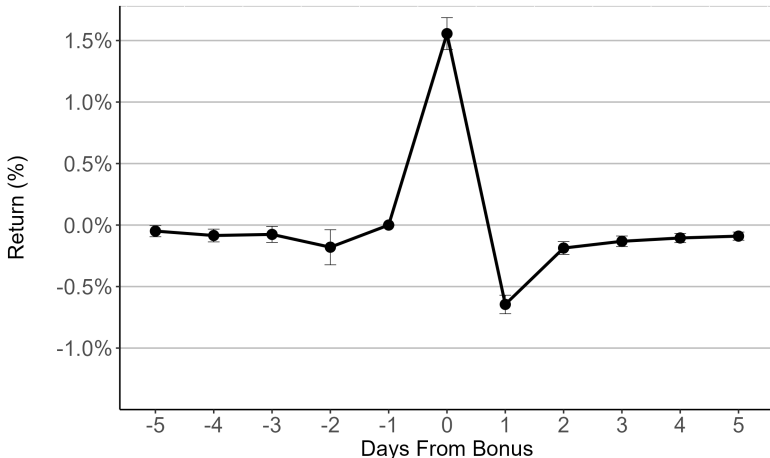
Higher bonus motivates greater trading around the HWM

# Differences-in-Differences Estimator

- Treatment = earning a bonus, which is
  - **Non-absorbing:** the HWM resets every January 1st
  - **Non-binary:** traders earn many bonuses a year
- Two-way fixed effects is biased with heterogeneous, staggered effects (Callaway and Sant'Anna 2021; Sun and Abraham 2021)
- Estimator of de Chaisemartin and D'Haultfoeuille (2026): change from the day before a bonus, vs. funds with the same treatment history not yet treated

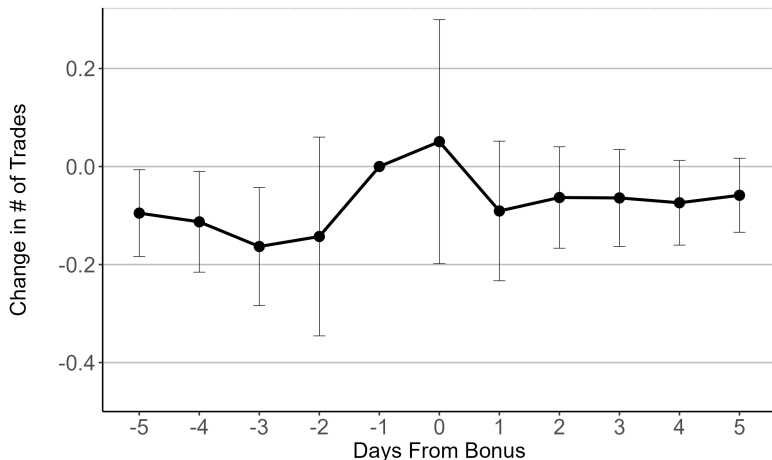
$$DID_{g,\ell} = Y_{g,F_{g-1}+\ell} - Y_{g,F_{g-1}} - \frac{1}{N_{g,F_{g-1}+\ell}} \sum_{g': D_{g',1}=D_{g,1}, F_{g'} > F_{g-1}+\ell} [Y_{g',F_{g-1}+\ell} - Y_{g',F_{g-1}}]$$

## Returns Around Bonuses



Traders earn a 1.6 pp higher return on the bonus day

## Number of Trades Around Bonuses

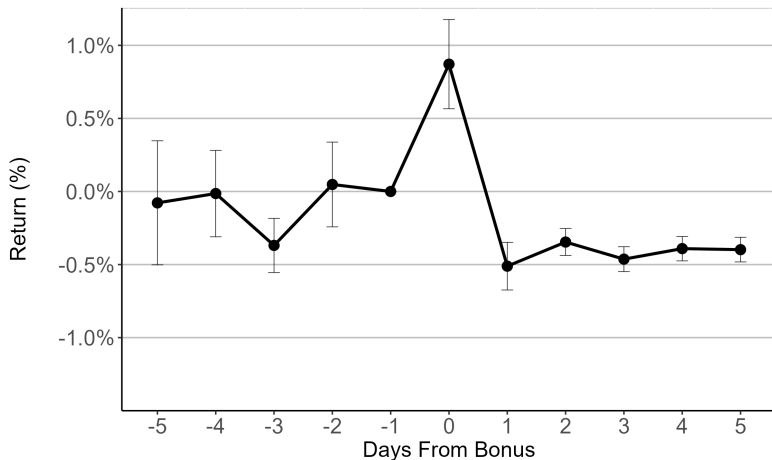


No significant increase in trades placed

# January Reset

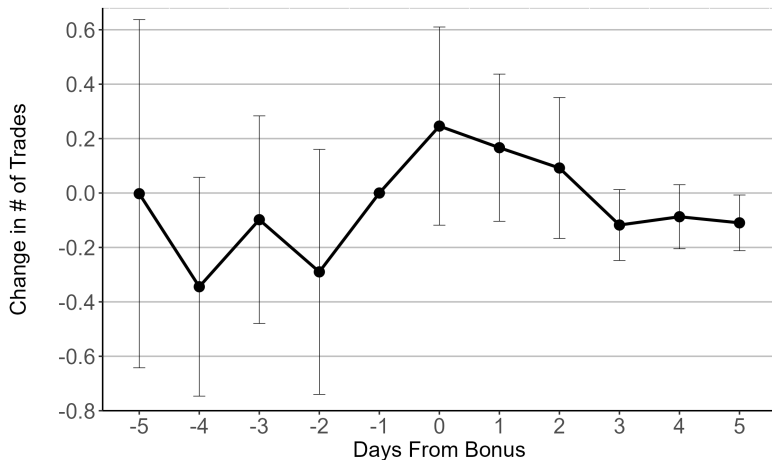
- Concern: more skilled traders may cross earlier or more often, and crossing depends on past performance
- Every January 1st all HWMs reset to that day's close
  - More easily achievable bonus allows us to isolate its effect
- Re-run the DiD on January only, first 3 bonuses per trader (437,046 observations)

## January: Returns Around Bonuses



Similar pattern to the whole sample

## January: Number of Trades Around Bonuses

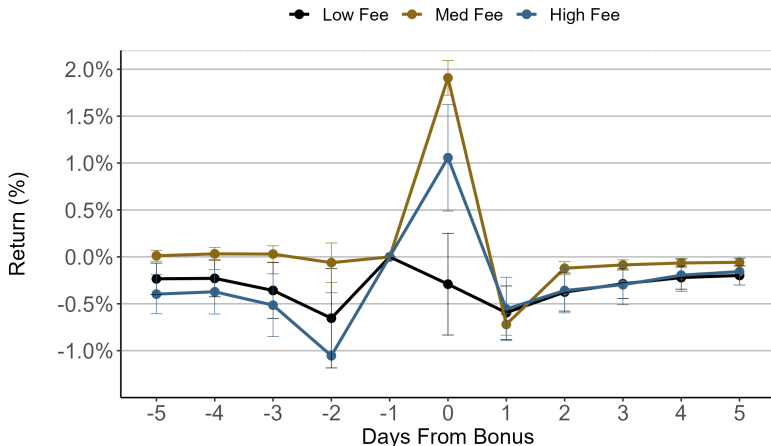


Noisy, yet positive, estimate of trades

# Who Pushes Hardest for the Bonus?

- If the bonus drives the return jump, traders with the most to gain should push hardest
- A higher bonus rate means a larger payout from crossing the HWM
- Test: re-run the DiD around bonuses separately for low, medium, and high fee traders
- Prediction: the bonus-day return jump shows up in the higher-fee groups

# Returns Around Bonuses By Performance Fee

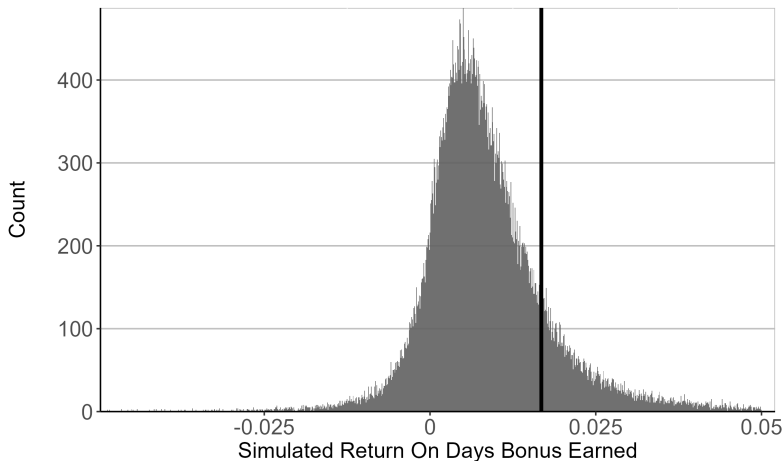


Higher fee traders have higher returns, consistent with incentives

# Luck or Strategy?

- Returns must be positive on a bonus day, so how big is a 1.6 pp jump?
- Luck vs. skill bootstrap (Kosowski et al. 2006; Fama and French 2010)
- For each bonus day:
  - Keep the fund's portfolio weights from the prior day
  - Draw that day's asset returns at random from the assets held by all funds crossing that day
  - Repeat 10,000 times
- Compare the simulated returns to the actual bonus-day return

# Bootstrap Return Distribution

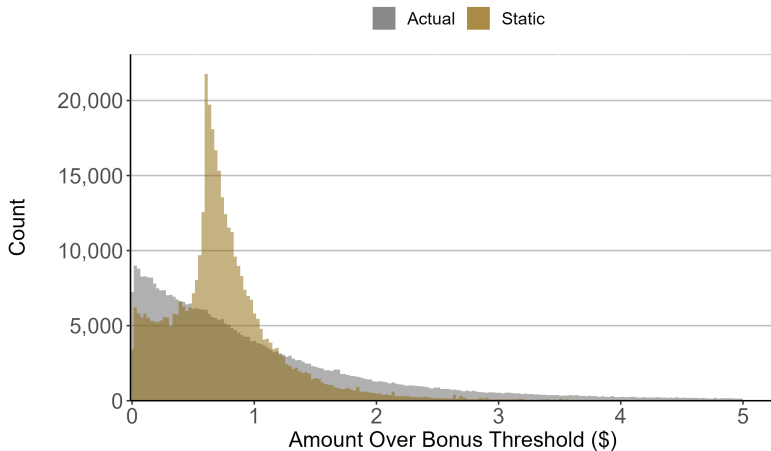


Observed 1.6% return is larger than 82% of simulations

# Static Portfolio Counterfactual

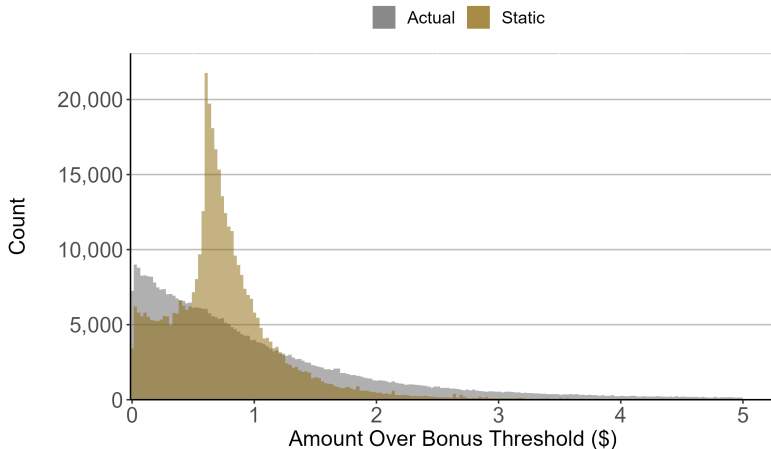
- What if traders did not trade at all?
- Freeze each fund's holdings on February 1st and hold them for a year
- Cumulative returns are similar and FF3 alpha is higher under a static setting for 63% of funds
- But is this optimal for bonus achievement?

## Distance Over Bonus Threshold



Under the static trading strategy 57% of fund-years achieve more bonuses

## Distance Over Bonus Threshold



Long right tail of actual crossings, consistent with HWM incentive

## Back of the Envelope

- Is chasing the bonus worth it for the trader?
- **Gain:** the bonus as a share of fund value,  $Bonus_{i,t}/Price_{i,t}$
- **Cost:** abnormal return, the next 10-day return after the bonus minus the 10-day return just before it

$$Net_{i,t} = \frac{Bonus_{i,t}}{Price_{i,t}} + \left( R_{i,t}^{+10} - R_{i,t}^{-10} \right)$$

- $Net_{i,t} < 0$ : the push for the bonus cost more than it paid
- Average net loss of 2.5% of fund value; median of -1.4%
  - Statistically different from zero; robust to window length and to overlapping bonus windows

# Summary of Additional Results

## ● Mechanism

[► Details](#)

- Assets traded on bonus days had higher past 10-day returns (sales 7.4% vs. 4.8% for purchases)
- Sold assets went on to earn more: 0.69 pp higher future returns than purchases
- Sales jump from about 40% to 48% of trades on bonus days

## ● Investors (new data, preliminary)

[► Details](#)

- Certificate trades rise about 25% on bonus days (top 20% most-traded)
- Volume rises too, but the estimate is noisy

# Conclusion

- Spike in returns corresponds with selling high-return assets and forgoing potential further gains
- High-frequency incentives can be counterproductive for retail investors
  - Exacerbating short-term biases and undermining sustainable performance
- Highlights importance of incentive frequency and interaction with agent type

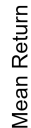
# Thanks!

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[www.jmohr.me](http://www.jmohr.me)

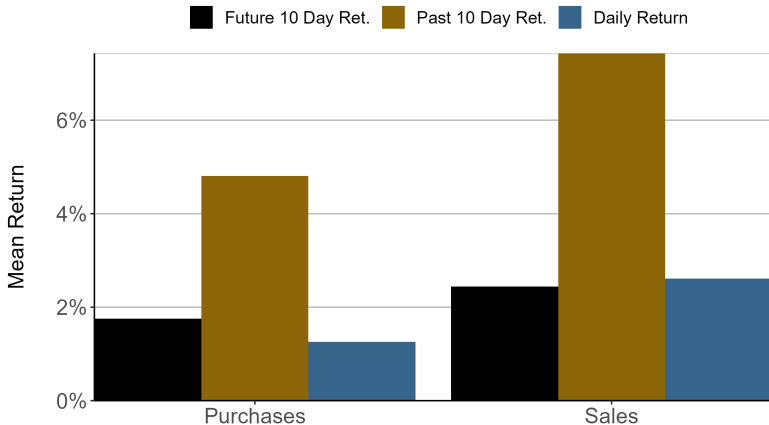
# Mechanism: What Do Traders Trade?

- Use transaction-level data
- Compare assets bought and sold on bonus vs. non-bonus days
- Look at each asset's past 10-day, same-day, and next 10-day return



[▶ Sales Table](#)
[▶ Purchases Table](#)

## Return Patterns: Purchases vs. Sales on Bonus Days

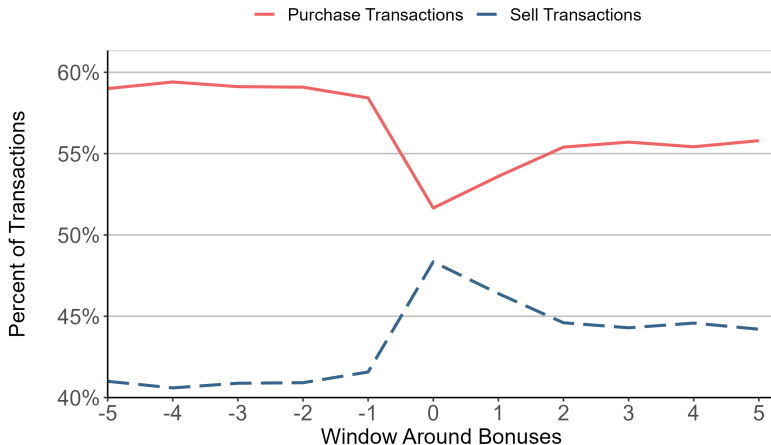


Future returns are larger for assets that traders sold on bonus days

## Selling to Lock In the Bonus

- Are traders selling recent winners to secure the bonus?
- Share of transactions that are sales, in the 10-day window around a bonus

# Buying vs. Selling Activity

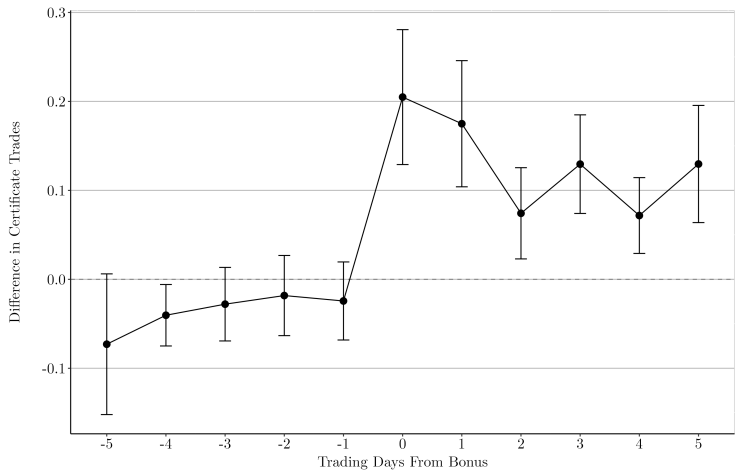


Consistent with selling recent winners to lock in the close price

# New Data: How Investors Trade the Certificates

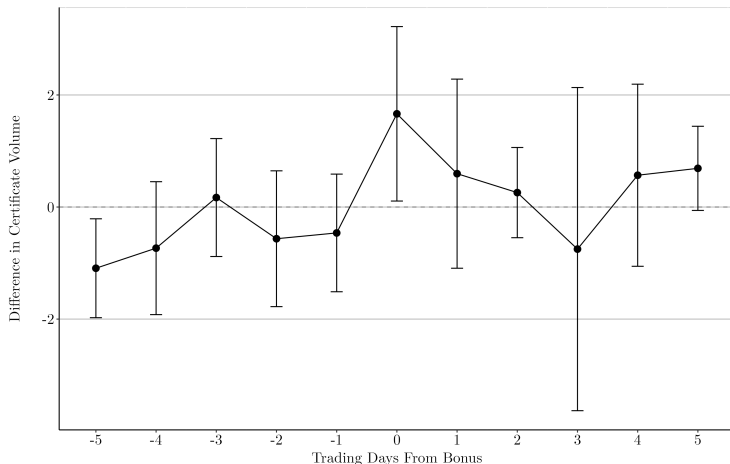
- Newly collected certificate trading data (preliminary results)
- Do investors respond to bonus days?
- Event study of certificate trades and volume around bonuses, in the most-traded certificates

## Certificate Trades Around Bonuses: Top 20% Most-Traded



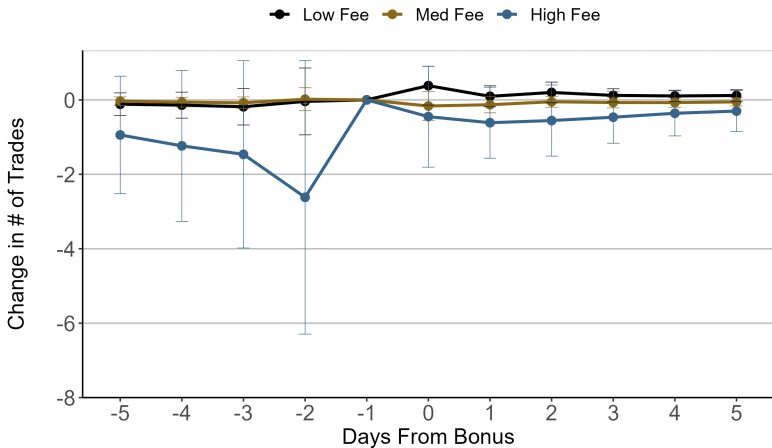
~25% of the 0.82 daily mean for these certificates

# Certificate Volume Around Bonuses



~23% of the 7.29 daily mean for these certificates

## Number of Trades Around Bonuses By Performance Fee



No difference across fee



## Bonus Achievement Difference in Differences

|              | Returns               | Number of Trades     | Volatility        |
|--------------|-----------------------|----------------------|-------------------|
|              | (1)                   | (2)                  | (3)               |
| Day -5       | -0.001**<br>(0.0002)  | -0.095**<br>(0.045)  | 0.003<br>(0.002)  |
| Day -4       | -0.001***<br>(0.0002) | -0.113**<br>(0.052)  | 0.004<br>(0.002)  |
| Day -3       | -0.001**<br>(0.0003)  | -0.163***<br>(0.061) | 0.006<br>(0.003)  |
| Day -2       | -0.002**<br>(0.001)   | -0.143<br>(0.103)    | -0.003<br>(0.004) |
| Day 0        | 0.016***<br>(0.001)   | 0.051<br>(0.127)     | 0.004<br>(0.004)  |
| Day +1       | -0.006***<br>(0.0004) | -0.091<br>(0.073)    | 0.001<br>(0.003)  |
| Day +2       | -0.002***<br>(0.0003) | -0.063<br>(0.053)    | 0.0002<br>(0.002) |
| Day +3       | -0.001***<br>(0.0002) | -0.064<br>(0.051)    | 0.001<br>(0.003)  |
| Day +4       | -0.001***<br>(0.0002) | -0.074<br>(0.044)    | -0.001<br>(0.002) |
| Day +5       | -0.001***<br>(0.0002) | -0.059<br>(0.039)    | -0.001<br>(0.002) |
| Observations | 3,250,182             | 3,250,182            | 3,250,182         |

## Transactions When Bonus Achieved: Sales

| Dependent Variables:                     | Next 10 Day Returns |                     | Previous 10 Day Returns |                     |
|------------------------------------------|---------------------|---------------------|-------------------------|---------------------|
|                                          | (1)                 | (2)                 | (5)                     | (6)                 |
| Bonus Achieved                           | 0.012***<br>(0.001) | 0.007***<br>(0.001) | 0.045***<br>(0.001)     | 0.021***<br>(0.001) |
| Sale Transaction                         | -0.002*<br>(0.001)  | -0.001<br>(0.001)   | 0.004**<br>(0.001)      | 0.005***<br>(0.001) |
| Bonus Achieved $\times$ Sale Transaction | 0.005***<br>(0.001) | 0.004***<br>(0.001) | 0.022***<br>(0.001)     | 0.020***<br>(0.001) |
| Previous 10 Day Returns                  | 0.129***<br>(0.003) | 0.121***<br>(0.003) |                         |                     |
| Daily Return                             |                     |                     | 0.022***<br>(0.005)     | 0.020***<br>(0.004) |
| <i>Fixed-effects</i>                     |                     |                     |                         |                     |
| Fund                                     | Yes                 | Yes                 | Yes                     | Yes                 |
| Date                                     |                     | Yes                 |                         | Yes                 |
| Observations                             | 3,620,016           | 3,620,016           | 3,635,974               | 3,635,974           |
| Adjusted R <sup>2</sup>                  | 0.026               | 0.021               | 0.025                   | 0.011               |

## Transactions When Bonus Achieved: Purchases

| Dependent Variables:                    | Next 10 Day Returns  |                      | Previous 10 Day Returns |                      |
|-----------------------------------------|----------------------|----------------------|-------------------------|----------------------|
|                                         | (3)                  | (4)                  | (7)                     | (8)                  |
| Bonus Achieved                          | 0.017***<br>(0.001)  | 0.011***<br>(0.001)  | 0.067***<br>(0.002)     | 0.041***<br>(0.001)  |
| Buy Transaction                         | 0.002*<br>(0.001)    | 0.000<br>(0.001)     | -0.004**<br>(0.001)     | -0.005***<br>(0.001) |
| Bonus Achieved $\times$ Buy Transaction | -0.005***<br>(0.001) | -0.004***<br>(0.001) | -0.022***<br>(0.001)    | -0.020***<br>(0.001) |
| Previous 10 Day Returns                 | 0.129***<br>(0.003)  | 0.121***<br>(0.003)  |                         |                      |
| Daily Return                            |                      |                      | 0.022***<br>(0.005)     | 0.020***<br>(0.004)  |
| <i>Fixed-effects</i>                    |                      |                      |                         |                      |
| Fund                                    | Yes                  | Yes                  | Yes                     | Yes                  |
| Date                                    |                      | Yes                  |                         | Yes                  |
| Observations                            | 3,620,016            | 3,620,016            | 3,635,974               | 3,635,974            |
| Adjusted R <sup>2</sup>                 | 0.026                | 0.021                | 0.025                   | 0.011                |